

Nano-chemistry

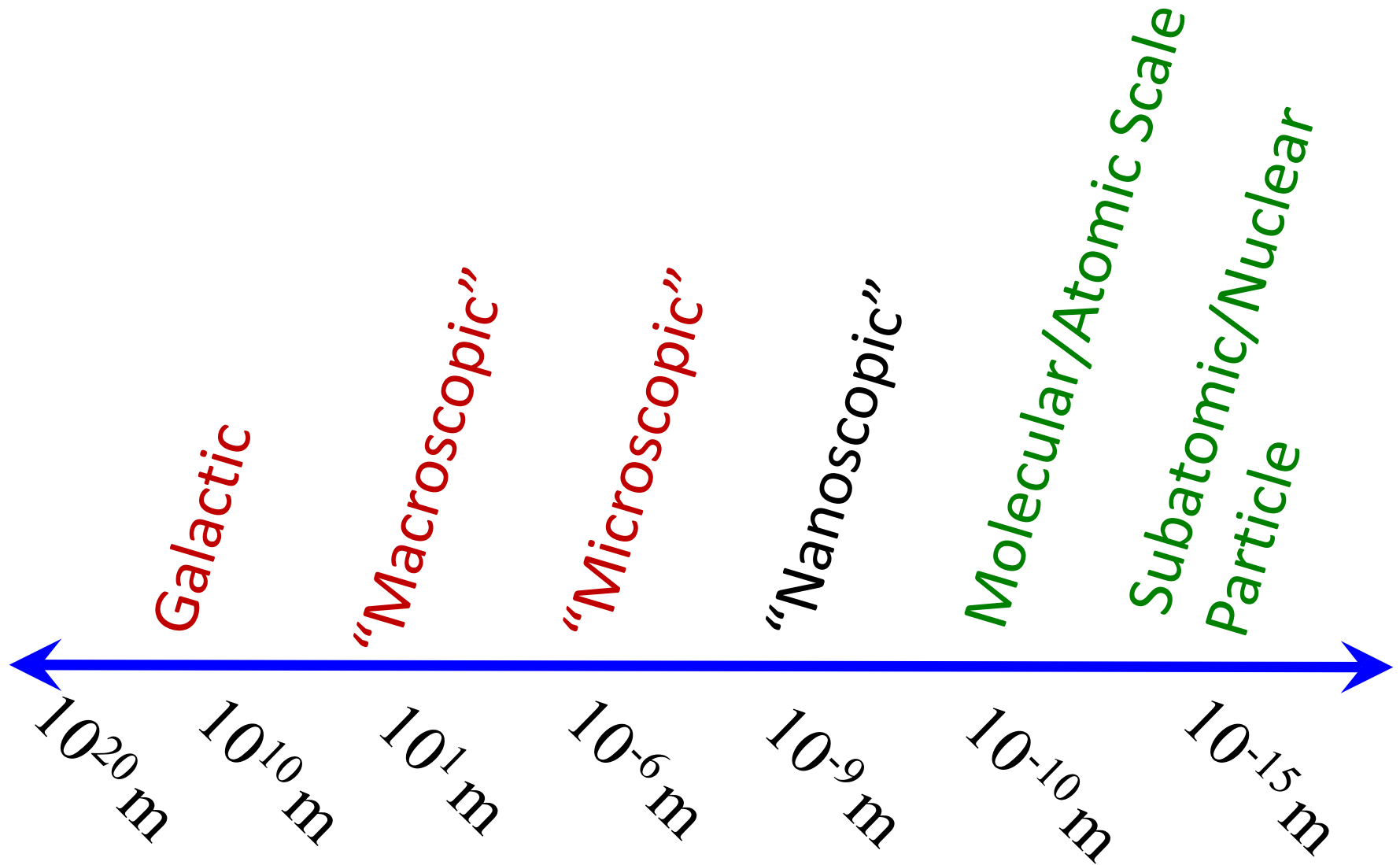
Fundamentals

Nanos [*Greek*]: the dwarf (*verb*, cause to seem small or insignificant in comparison)

One nanometer (nm) is one billionth, or 10^{-9} of a meter. The comparative size of a nanometer to a meter is the same as that of a marble to the size of the earth .



Nano: The Middle Ground





A human fingernail grows
1 nanometre every second.



A man's beard grows
5 nanometres every second.

Nanoscience, Nanochemistry And Nanotechnology

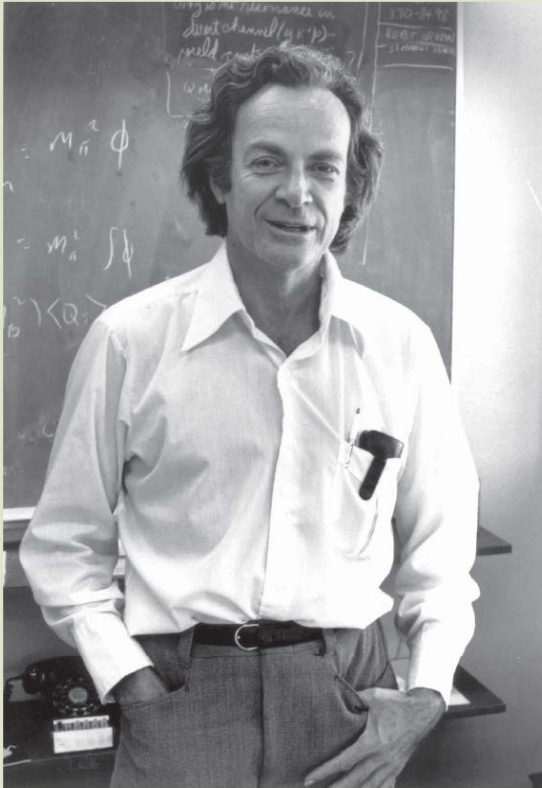
Nanoscience – the study of matter at very small length scales: 100 nanometres and below.

Nanochemistry – the synthesis of nanoscale building blocks of different size, shape, composition, surface structure, charge and functionality.

Nanotechnology – the products that are derived from this knowledge

PIONEERS OF NANOTECHNOLOGY

FIG. 1.0 Nobel Prize laureate Richard Feynman at the blackboard.



The ideas and concepts behind nanoscience and nanotechnology started with a talk entitled “**There’s Plenty of Room at the Bottom**” by physicist **Richard Feynman** at an **American Physical Society** meeting at the California Institute of Technology in **1959**, long before the term nanotechnology was used. In his talk, Feynman described a process in which scientists would be able to manipulate and control individual atoms and molecules. It wasn't until **1981**, with the development of the **scanning tunneling microscope** that could "see" individual atoms, that modern nanotechnology began.

He shared the **Nobel Prize for Physics** in **1965** with **Schwinger** and **Tomonaga** for quantum electrodynamics.

PIONEERS OF NANOTECHNOLOGY

“Why cannot we write the 24 volumes of Encyclopaedia Britannica on the head of a pin?”

Atoms on a small scale behave like nothing on a large scale.....
So, as we go down and fiddle around with atoms down there, we are working with different laws....”

Feynman covered various aspects of nanoscience and technology in his lecture:

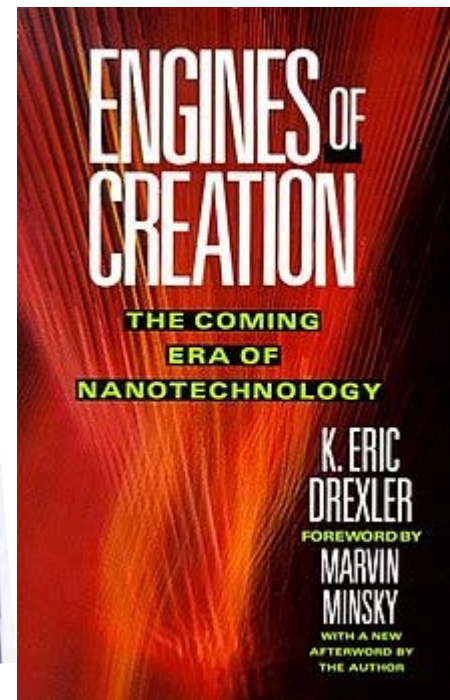
- ❖ the need for electron microscopes,
- ❖ miniaturising the computer,
- ❖ using evaporation as a means of miniaturising,
- ❖ developing miniature tools to manipulate nanoobjects at the atomic level (a master-slave system).

Grey goo

Eric Drexler (born April 25, 1955) , American engineer

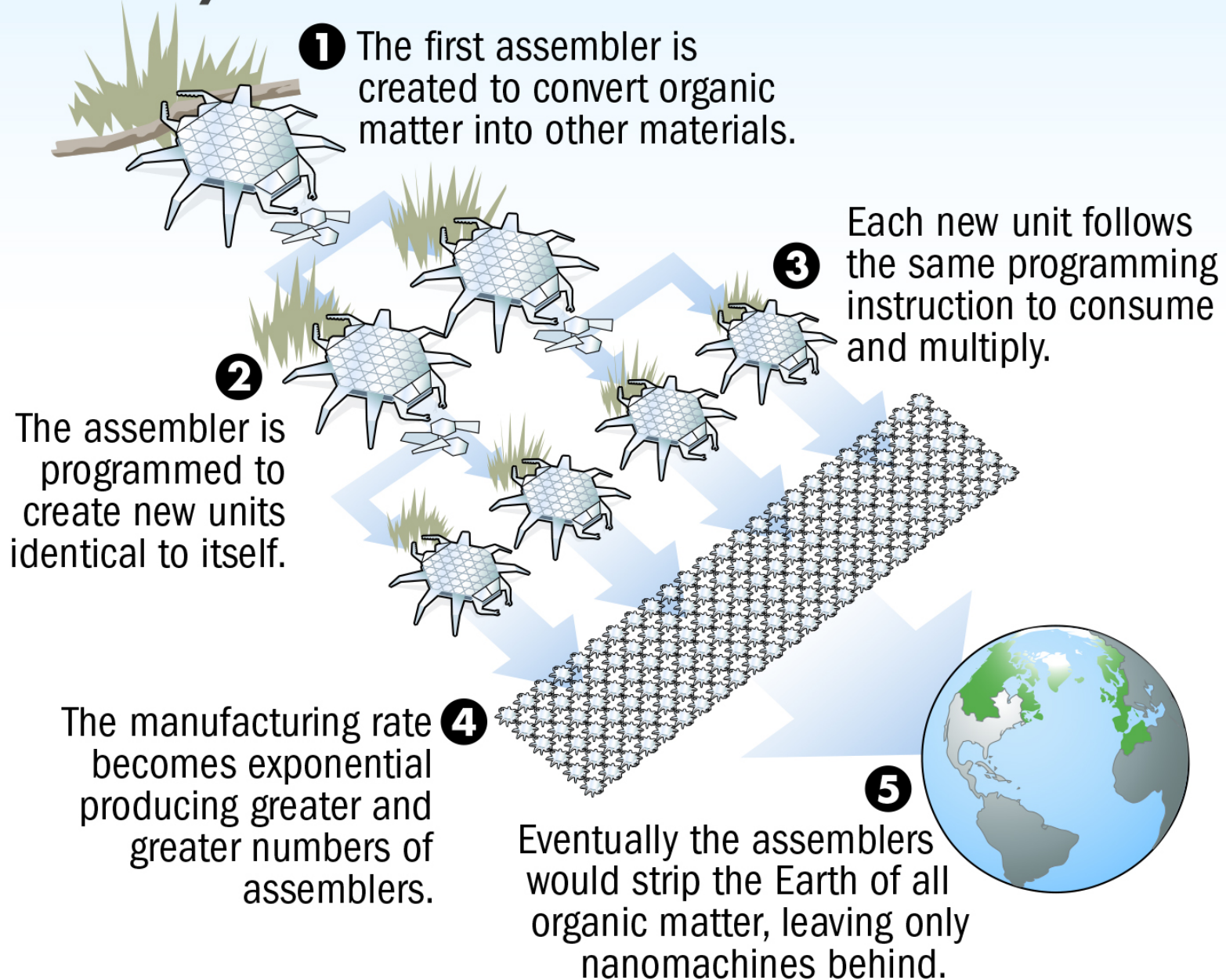
In his famous book "Engines of Creation: The Coming Era of Nanotechnology" published in 1986, Drexler described a world completely transformed by molecular assemblers. He was convinced that they would be able to build anything with absolute precision and no pollution. They would even enable the colonization of the solar system. Drexler was also aware that these "assemblers" would pose serious threats to life on Earth.

He coined the word "**Grey goo**" a hypothetical situation in which out-of-control molecular assemblers replicating themselves would consume all living matter on Earth. He, therefore, urged that the society must thoroughly understand both the creative and the destructive scope of these assemblers and make sure to develop safety mechanisms to ensure only the beneficial applications. However, Drexler's prophecy of molecular assemblers becoming a reality has not found acceptance.



The Grey Goo Scenario

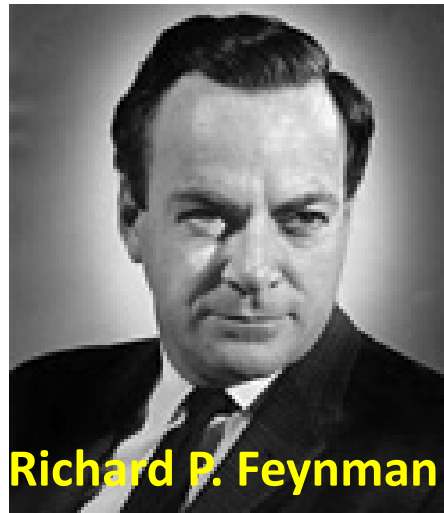
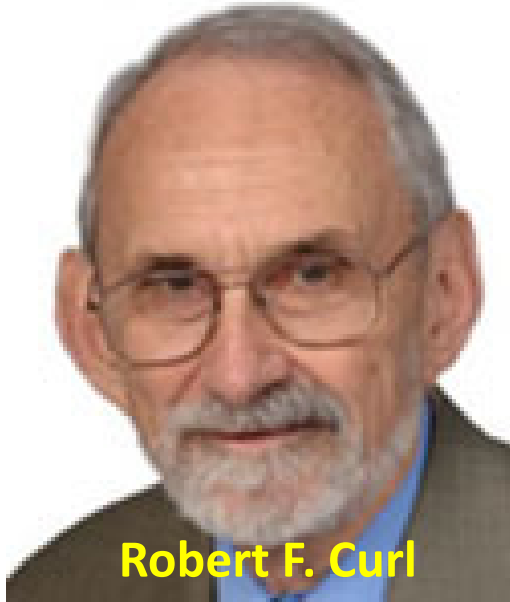
©2010 HowStuffWorks



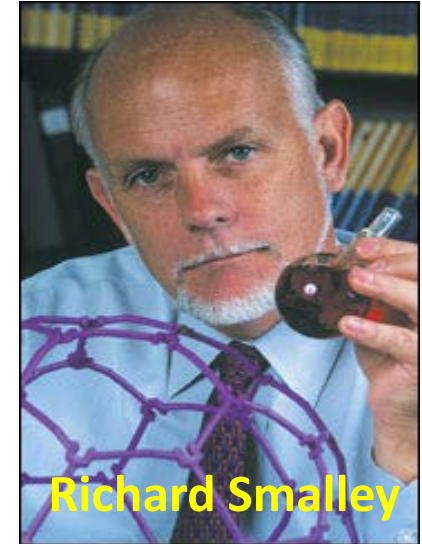
Nanotechnology Pioneers

May 11, 1918 – February 15, 1988

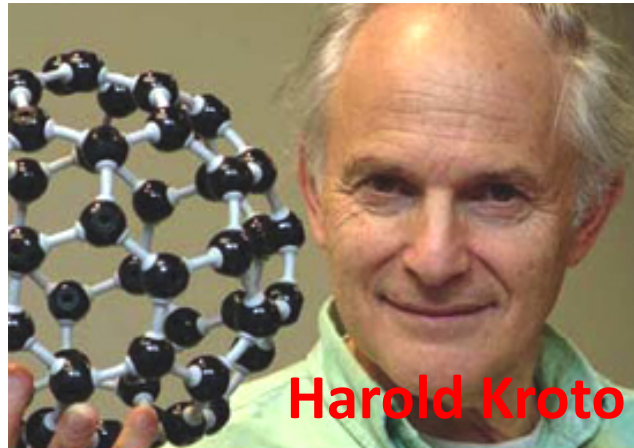
August 23, 1933 – July 3, 2022



June 6, 1943 – October 28, 2005

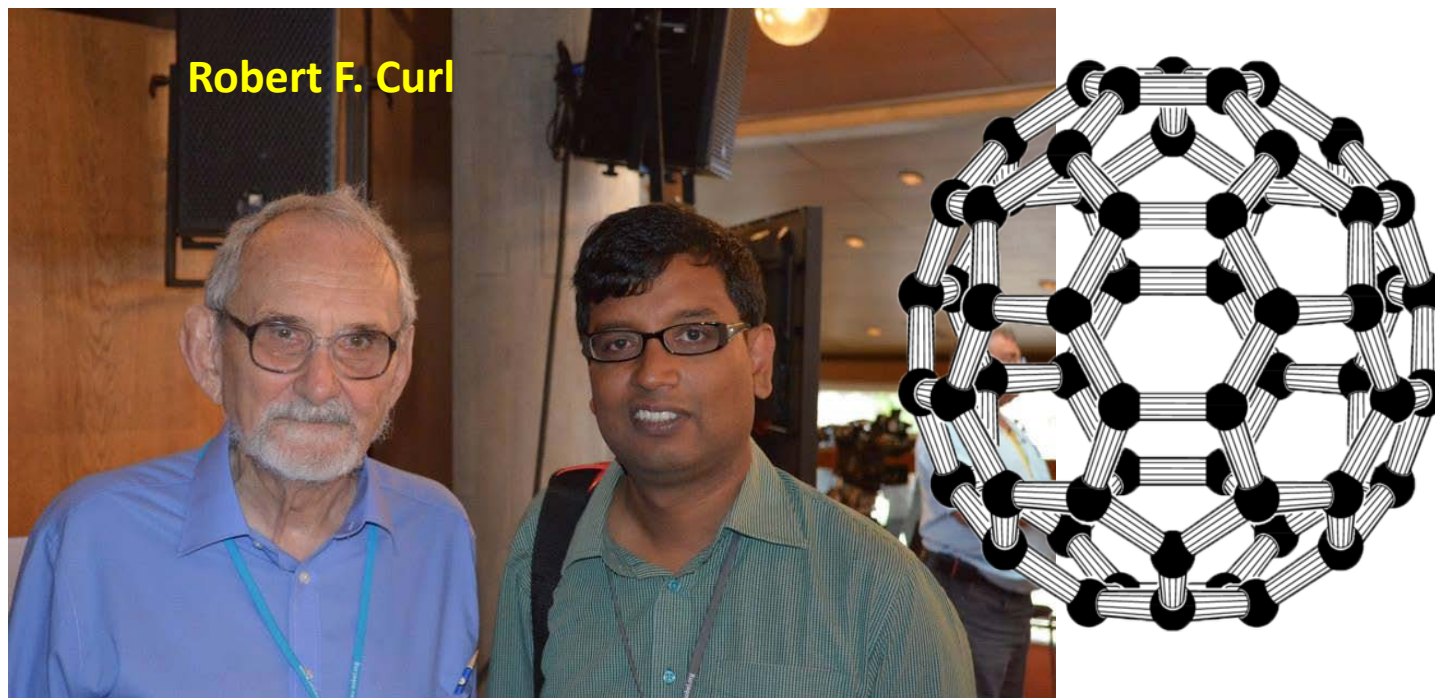


7 October 1939 – 30 April 2016



Nobel Prize in Chemistry in 1996 for the discovery of the nanomaterial buckminsterfullerene

63rd Meeting of Nobel Laureates (Chemistry) – 2013: Germany



Nobel Prize in Chemistry in 1996 for the discovery of the nanomaterial **buckminsterfullerene**, and hence the fullerene class of materials, along with [Richard Smalley](#) (also of [Rice University](#)) and [Harold Kroto](#) of the [University of Sussex](#).

Professor of Chemistry at Rice University

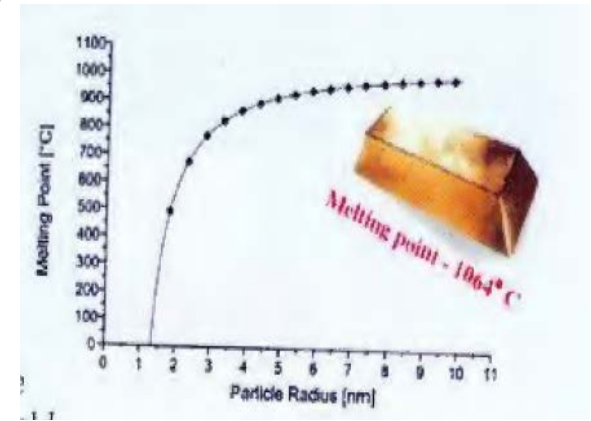
Why nano?

At the nanoscale, strange things happen to materials –
their **properties can change**.

Reactivity - As particles get smaller they tend to react differently with their environment than larger particles.

Size - Smaller particles can have different optical properties: their colours change because different sizes of particle reflect and absorb light differently.

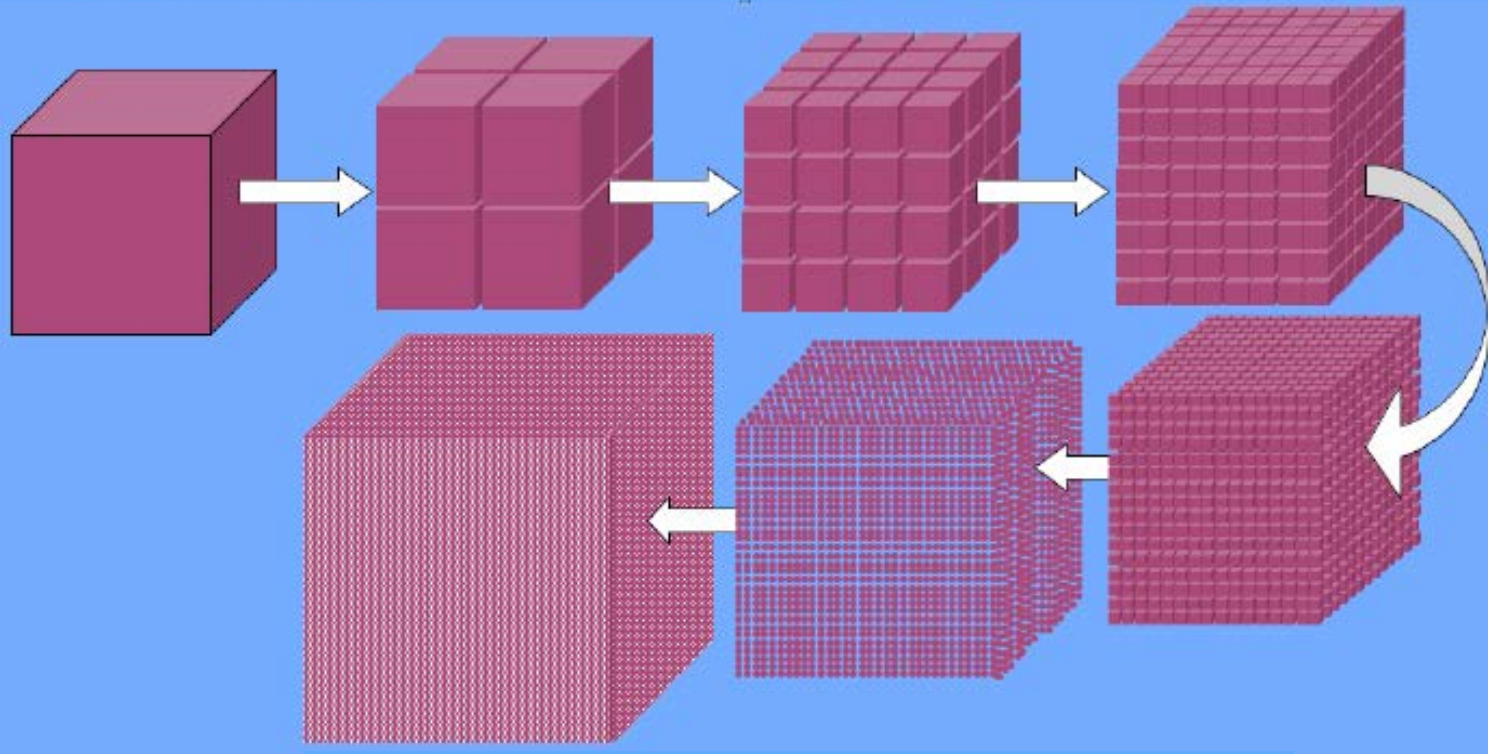
Magnetism - Smaller particles can have different magnetic properties than larger.



Wavelike (quantum mechanical) properties of electrons inside matter and atomic interactions are influenced by materials variations on the nanometer (nm) scale. **By creating nanometer-scale structures, it is possible to control the fundamental properties of materials**, such as their **melting temperature**, **magnetic properties**, **charge capacity**, and **even their color**, without **changing the materials' chemical compositions**. Making use of this potential will lead to new, high-performance products and technologies that were not possible before.

Collective Surface Area

Each time the material is divided, the surface area increases geometrically.



As materials get smaller and smaller.... their collective surface area gets larger and larger.

Collective surface area:

$$A_{collective} = \sum A_{particles}$$

Singular Surface Area and the Surface-to-Volume Ratio

When we refer to singular surface area, we refer to the surface area of a single nanoparticle.

Area is a function of a dimension squared, e.g. d^2 , x^2 or r^2 .
Volume is a function of that dimension cubed, e.g. d^3 , x^3 or r^3 .

Surface-to-volume ratio (d^2 / d^3) scales as the inverse of the dimension: e.g. d^{-1} . In other words, as the dimension approaches smaller and smaller limits, the surface-to-volume ratio scales as the inverse of the dimension.

Surface-to- volume ratio increases with the decrease in particle sizes

Nano in History



- Lycurgus Cup (gold-silver nanoparticles in NaO glass)
- Damascus sword (carbon nanotubes in steel)
- Medieval glasses (metal and semiconducting nanoparticles)
- Catalyst particles for making soap (ash, clays)
- Clay for bricks (clay is a nanomaterial!)



(a)



(b)

Source: Images reprinted with permission from British Museum Images.

FIG. 1.8

Two versions of the famous Lycurgus Cup, displayed in the British Museum, are shown in this figure. (a) The green appearance is produced by reflection. (b) Red and purple colors are due to transmitted light. The apparent dichroism is due to the interaction of light with gold-silver and copper nanometals and alloys embedded within the soda glass matrix of the cup. King Lycurgus is shown being strangled by the maenad Ambrosia, who was changed into a vine by the goddess Diana. The sad state of King Lycurgus, a Thracian king in 800 B.C., was a consequence of his assault on the god Dionysius and Ambrosia.

Dichroism

“Dichroism” means “two colors”. Indeed, 2 colors are revealed by the Lycurgus Cup—
depending on your point of view:

1. Green on reflection
2. Red upon transmission

THERE ARE SEVERAL REASONS FOR THE APPARENT DICHROISM:

1. Metal particle size range from 40 nm to 100 nm
 2. Contributions of gold / silver / copper particles
 3. Absorption and scattering
- The larger silver nanoparticles dominate the scattering (i.e. mini-reflections) that results in the green appearance.
 - The smaller gold nanoparticles are responsible for the ruby color on transmission.
 - Alloys of the three metals may play a more subtle role

The Damascus Sword

- The Damascus Sword (ca. 10th to 18th centuries) a composite metal and the first technological (and unwitting) use of carbon nanotubes, other carbon materials, cementite nanowires and glasses
- The Damascus sword was strong, hard, flexible and very sharp— due to its laminar structure, the result of its special forging process
- Carbon nanotubes were formed from a special iron (steel) made in India called “Wootz steel”, a high carbon and pure steel.



Peter Paufler at the Technical University in Dresden, Germany and his colleagues studied samples from the blade of the Damascus sword made in the 17th century.

The blade of Damascus sword could **cut a piece of silk** as it fell to the ground and at the same time retain their sharpness even after cutting through **stone or metal or other swords**.

They found tiny nanowires and nanolubes in the sample.



Dagger made from wootz steel



Complex surface patterns
in a Damascus sword

Seeing at the Nanoscale

How do scientists see what's going on in the extremely small world of nanotechnology?

- ❑ The microscopes that are typically used in high school and college won't do the job.
- ❑ Nanoscientists use high-powered microscopes that use unique methods to allow them to see the **surface features on the atomic scale**, effectively opening the door to modern nanotechnology.

Nanotools

There are several breakthroughs in instrumentation and fabrication that brought on the Nano Age.

Scanning electron microscope (SEM)

Transmission electron microscope (TEM)

Scanning tunneling microscope (STM)

Atomic force microscope (AFM)

Surface enhanced Raman spectroscopy

Lithography

Nanoimprint lithography

Nanosphere lithography

Dip-pen lithography

Nano fabrication

The fabrication side has seen the emergence of two paradigms, respectively referred as “top-down” and “bottom-up”.

The **top-down method** begins with large homogeneous objects and removes material as needed to create smaller-scale structures. **Similar to the work of a sculptor in carving a face from a block of marble**, this approach is epitomized by lithography techniques, the cornerstone of microelectronics fabrication.

A **bottom-up approach** involves **putting together smaller components (such as individual atoms and molecules) to form a larger and more complex system by leveraging naturally occurring chemical, physical, and biological processes.**

Different methods of preparing nanomaterials

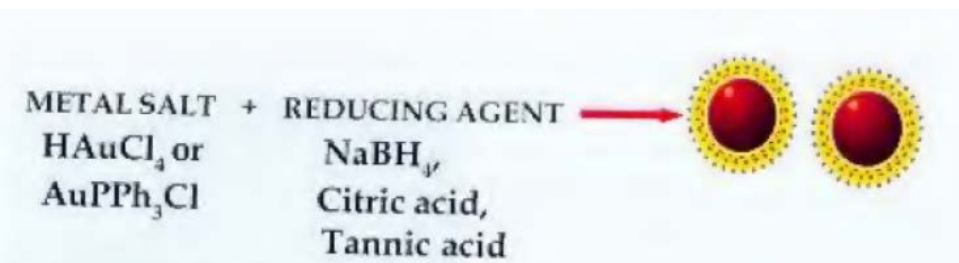
Physical methods: We can grind materials using ball mills or evaporate materials to the gas phase to obtain small particles. We can also use lasers to produce small particles.

Hydrothermal and solvothermal methods

- ❖ When water is used as the boiling solvent, the method is called the hydrothermal method.
- ❖ When a boiling organic solvent like a hydrocarbon is used, it is called the solvothermal method.
- ❖ For example, a metal acetate heat in a bomb in a hydrocarbon solvent, we get metal or metal oxide nanoparticles,
- ❖ Solvothermal reactions are popular for the synthesis of various kinds of nanomaterials. One often adds a polymer or a capping agent to the reaction mixture to control the shape and size of particles. After synthesizing the particles, one can also carry out selective crystallization to get particles of the same size."

Different methods of preparing nanomaterials

Take 23,5 ml of 6.38 mM NaOH solution in a round bottom flask containing a magnetic pellet and stir the solution. (We have to add PVA, a polymer to control the size of the particles), using a micropipette add 0.5 ml of a freshly prepared 50 mM solution of THPC [tetrakis (hydroxymethyl) phosphonium chloride], This is used as a reducing agent. Note that the solution is colorless. Now, slowly add 1 ml of aqueous solution of chloroauric acid, HAuCl_4 , a gold compound, The color changes immediately. The solution containing chloroauric acid becomes pale brown (see (b)). The color is due to the formation of nanocrystals of gold.



Nanocomposite

Composites are generally two or more closely interacting materials in intimate contact with each other. Intimate contact can be down to the atomic or molecular level. Polymer science uses the concept of a composite as a material formed by combining different phases.

There are many ways of combining phases and these are: filling, blending, compounding, mixing, melting and assembling.

Composite materials are solid ones with multiple phases, which is a combination of two or more materials with different physical and chemical properties.

In composite materials, one phase is usually continuous and called the matrix, while the other phase is a reinforcing material called the dispersed phase.

Nanocomposite

Nanocomposites are a new class of composites that have a dispersed phase with at least one ultrafine dimension, typically a few nanometers. Nanocomposites possess special properties not shared by conventional composites, due primarily to large interfacial area per unit volume or weight of the dispersed phase (e.g., 750 m²/g).

Clay layers dispersed at the nanoscale in a polymer matrix act as a reinforcing phase to form polymer–clay nanocomposites, an important class of organic–inorganic nanocomposites.

These nanocomposites are also referred to as polymer–silicate nanocomposites and organic–inorganic hybrids.

Polymer–clay nanocomposites can drastically improve mechanical reinforcement and high-temperature durability, provide enhanced barrier properties, and reduce flammability

The classification of nanocomposites

Classification 1. This classification is based on the nanomaterial's dimensional (D) morphology.

- (a) 0-D nanopowder or nanoparticle;
- (b) 1-D nanowire, nanowisp, nanosilk, nanotube or nanocrystalline whisker;
- (c) 2-D nanomaterial with morphology of layer, lamellar or belt structure;
- (d) 3-D nanomaterial with morphology of cylinder, solid or block structure.

The classification of nanocomposites

Classification 2. This classification is based on polymer morphology, shape or state in the nanocomposites, in which the inorganic particle is a 0-D nanoparticle.

- (a) 1-D nanocomposite when the polymer in the composite is in solution;
- (b) 2-D nanocomposite when the polymer in the composite is in film or membrane shape;
- (c) 3-D nanocomposite when the polymer in the composite is in a powder state or morphology;
- (d) 0-D nanocomposite formed with two different nanopowders.

The classification of nanocomposites

Classification 3. This classification is based on a nanophase of tube- or non-tubelike material in the nanocomposites.

- (a) *Tube-like*, e.g., single-wall carbon nanotube, TiO_2 Tube, or Al_2O_3 tube.
- (b) *Non-tube-like*, e.g., layer of clay, lamellar, silk, belt or band-like shape.

Polymer-clay nanocomposite

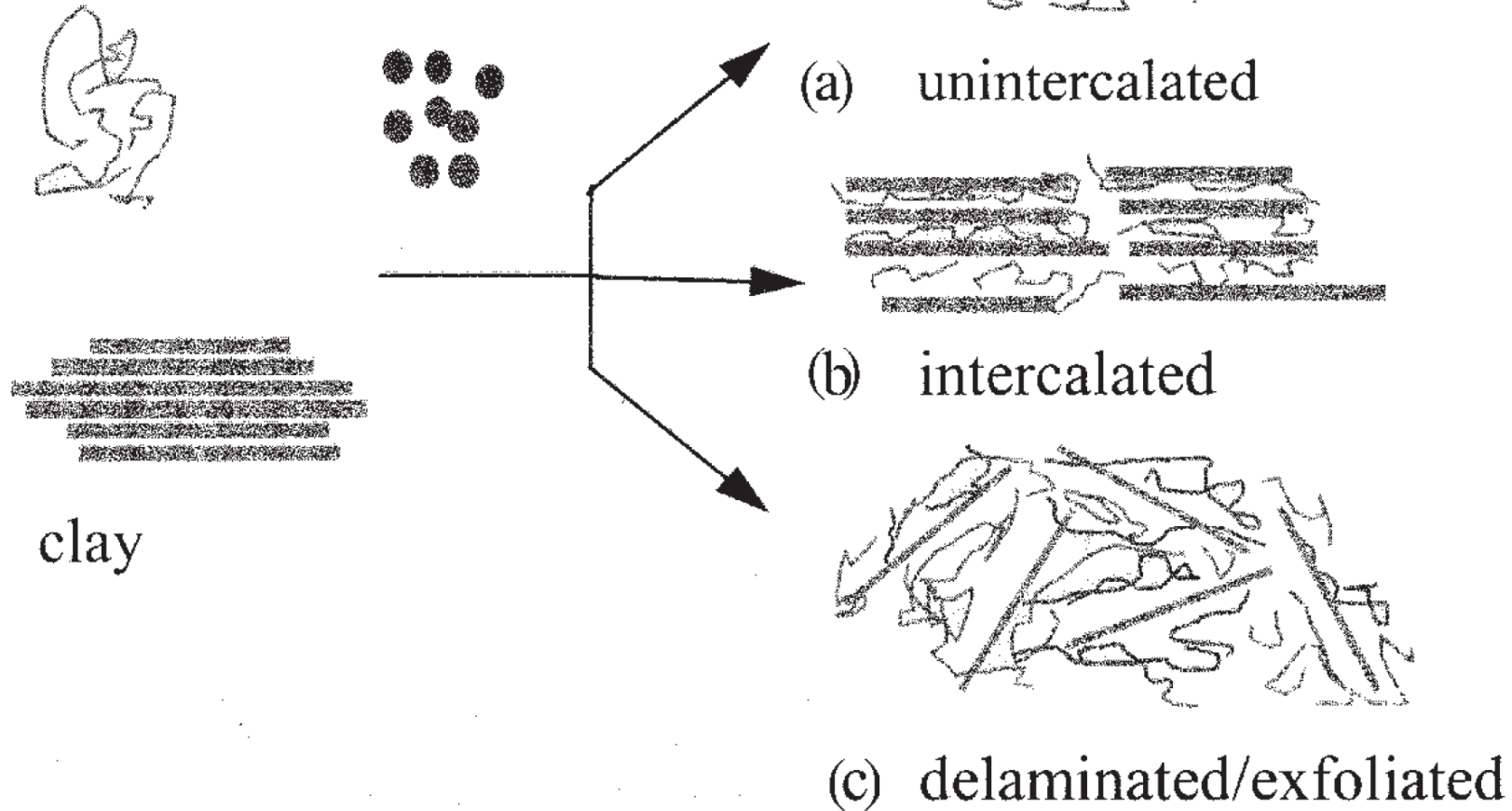
Colloid and surface chemistry play important roles in the synthesis of polymer–clay nanocomposites. Dispersion of clay layers in polymers is hindered by the inherent tendency to form face-to-face stacks in agglomerated tactoids due to high interlayer cohesive energy.

Nanoscale dispersion of the clay tactoids into individual nanolayers is known as exfoliation or delamination. Exfoliation is further prevented by the incompatibility between hydrophilic clay and hydrophobic polymers.

Treatment or functionalization of clay by adsorption of organic molecules weakens the interlayer cohesive energy. Intercalation, i.e., penetration of organic molecules into the clay interlayers, increases the compatibility between clay and polymer matrix.

Due to the negative charge on the clay surface, cationic surfactants and polymers are commonly used for intercalation. The ion exchange of inorganic cations in clay galleries by organic cations renders the clay organophilic.

Polymer or Monomer



Schematic illustration of three types of polymer–clay composites.

Polymer-clay nanocomposite

- a) The first type is a conventional composite that contains clay tactoids with the nanolayers aggregated in un-intercalated face-to-face form. In this case, the clay tactoids are dispersed simply as a segregated phase, resulting in poor mechanical properties of the composite.
- b) The second type is intercalated polymer–clay nanocomposite, which is formed by the infiltration of one or more molecular layers of polymer into the clay host galleries.
- c) The last type is exfoliated polymer–clay nanocomposites, characterized by a low clay content, amonolithic structure, and a separation between clay layers that depends on the polymer content of the composite. Exfoliation is particularly desirable for improving specific properties that are affected by the degree of dispersion and resulting interfacial area between polymer and clay nanolayers.

Polymer-clay nanocomposite

When the inorganic cations are exchanged by organic cations, such as from surfactants and polyelectrolytes, the clay surface changes from hydrophilic to hydrophobic or organophilic.

The organic cations lower the surface energy and decrease the cohesive energy by expanding the interlayer distance, thus facilitating the wetting and intercalation of monomer or polymer.

The organic cations may contain various functional groups that react with monomer or polymer resin to improve interfacial adhesion between clay nanolayers and polymer matrix.

How reinforcement occurs in polymer-clay nanocomposite?

Homogeneous dispersion of clay nanolayers in a polymer matrix provides maximum reinforcement via distribution of stress and deflection of cracks resulting from an applied load.

Interactions between exfoliated nanolayers with large interfacial area and surrounding polymer matrix lead to higher tensile strength, modulus, and thermal stability.

Conventional polymer–filler composites containing micron-size aggregated tactoids also improve stiffness, but at the expense of strength, elongation, and toughness.

Factors influencing the properties of polymer–clay nanocomposites

The formation of polymer– clay nanocomposites or hybrids is affected by

- the packing density and
- chain length of the surfactant in the organoclay,
- the charge density of the clay,
- specific groups on the polymer,
- and the type of bonding at the polymer/clay interface (i.e., hydrogen, dipole–dipole, van der Waals, or covalent).

Preparation and technology for nanocomposites

In nanocomposite materials, particle dispersion is one of the most urgent problems to be addressed.

(1) Sol–gel method. The treating agent is added to the nanoparticle precursors, forming a sol by a hydration process. The sol forms a surface layer on the particle after recovering the gel. Then some monomers of solvent-soluble (e.g., polyvinylpyrrolidone (PVP)) or water-soluble (e.g., polyvinyl alcohol (PVA)) polymers are introduced; nanocomposites are prepared in one pot. This method has a narrower selecting space for organic polymers and nanomaterials, but is very popular for nanocomposites because the polymer is soluble in the media.

(2) Nanoprecursor methods. Nearly all of the oxide compounds or their particles can be transformed into superfine powders. Some have complicated compositions not suitable for the preparation of nanocomposites or composite nanoparticles. These materials are transformed into an intermediate form, called the precursor, in which the homogeneous raw material with extremely high purity is obtained.

Preparation and technology for nanocomposites

(3) Intercalation method. For layered compounds or silicates, agents or treating reagents are intercalated into the layer space. These intercalates are nanomaterial precursors with controlled layer morphology and are exfoliated during the polymerization reaction.

(4) Blending methods. Nanoparticles are prepared via a sol–gel process and are blended with treating agents such as siloxane, surfactants, lower molecular weight molecules, oligomers or polymers. By blending or mixing, particles covered with the treating agents are obtained. When a treating agent is a polymer, a core (inorganic)–shell (polymer cover) structure is usually obtained. The nanoparticles are not directly mixed with polymers but with an intermediate material.

(5) Filling methods. The composite nanoparticles or nanocomposites are prepared by filling organic molecules or polymers into inorganic particle pores, surfaces and interfaces by means of mechanical powder mixing, liquid dissolving or heat melting. This method is versatile and has wide applications for either treating or preparing nanomaterials or nanocomposites.

Quantum dots

Quantum dot A particle that exhibits a size quantization effect in at least one dimension.

Semiconductor quantum dots are signify a class of materials in which quantum confinement effects are investigated in greater detail. They are also referred to as 'semiconductor nanocrystals'.

Quantum dots have properties combined between Those of bulk semiconductors Those of atoms

Very small semiconductor particles with a size comparable to the Bohr radius of the excitons (separation of electron and hole).

Typical dimensions: 1 – 10 nm

Different shapes (cubes, spheres, pyramids, etc.)

Quantum confinement

Quantum confinement refers to the phenomenon in which the movement of **electrons and other charge carriers in a material is restricted to certain discrete energy levels due to the material's small size or confinement in all three dimensions.**

This confinement leads to quantized energy levels.

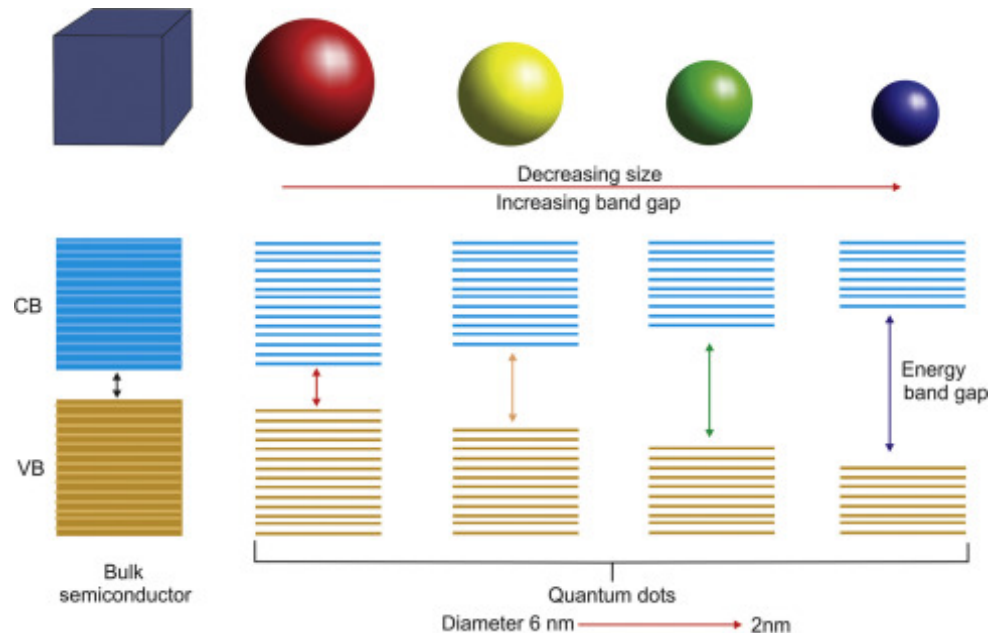
When the size of a **semiconductor material becomes comparable to or smaller than the characteristic length scale of its electrons (called the "Bohr radius"), quantum confinement effects become significant.**



CdSe quantum dot

Quantum confinement

- In bulk semiconductors, electrons can have a range of energy levels, forming continuous energy bands. However, when the semiconductor is confined to a small size, **the available energy levels become quantized, leading to the formation of discrete energy levels** called energy "quantum dots."
- These quantized energy levels give rise to unique optical and electronic properties that are different from those of bulk materials.
- The size of the quantum dot determines the energy gap between its electron energy levels. As the size decreases, the energy gap between the levels increases.



The most common methods of manufacturing semiconductor quantum dots are colloidal growth, epitaxial growth, and ion implantation.

Quantum Dots

A periodic table of elements with a focus on semiconductor materials. Elements are color-coded: Phosphorus (P) is light blue, Arsenic (As) is green, and Indium (In) is red. A black line connects Phosphorus (P) to Indium (In). The table includes element symbols, atomic numbers, and names. The title 'Quantum Dots' is in the top left corner. A video player interface is at the bottom, showing a progress bar at 1:52 / 4:00 and a CC logo.

			boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
			aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
unnilunium 110 Uu	unnilunium 111 Uu	unnilunium 112 Uu						

Applications of Quantum dots

- Photovoltaic devices: solar cells
- Biology : biosensors, imaging
- Light emitting diodes: leds
- Quantum computation
- Flat-panel displays
- Memory elements
- Photodetectors
- Lasers



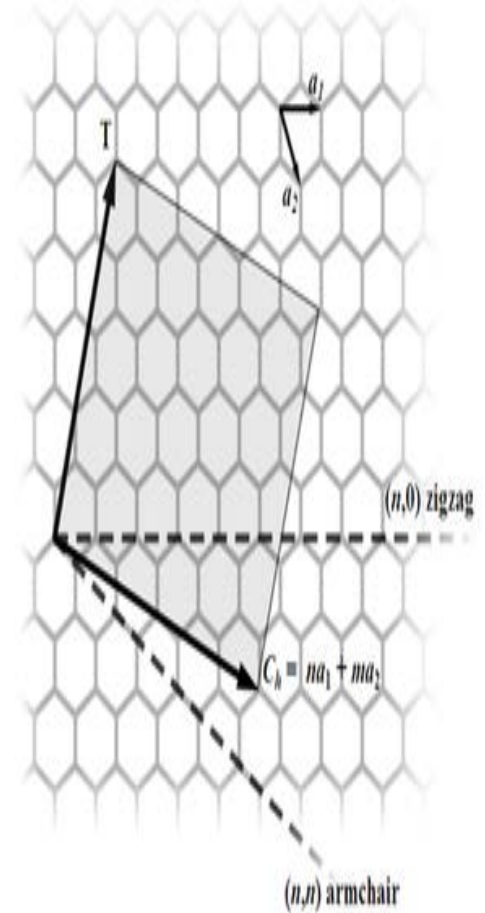
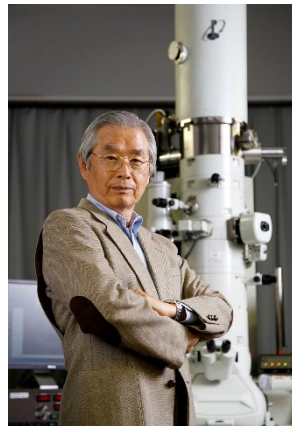
Cadmium-free quantum dots “CFQD”

- ❖ In many regions of the world there is now, or soon to be, legislation to restrict and in some cases ban heavy metals in many household appliances such as IT & telecommunication equipment, Lighting equipment , Electrical & electronic tools, Toys, leisure & sports equipment.
- ❖ For QDs to be commercially viable in many applications they **MUST NOT CONTAIN cadmium or other restricted elements LIKE mercury, lead, chromium.**
- ❖ So research has been able to create non-toxic quantum dots using [silicon](#).

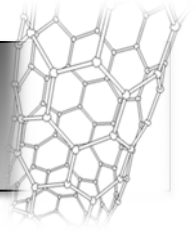
What are carbon nanotubes.

Carbon nanotubes (cnts) are allotropes of carbon with a cylindrical nanostructure. Its length-to-diameter ratio of up to 132,000,000:1, significantly larger than for any other material.

The history of carbon nanotubes starts from 1991 by Japanese researcher **Sumio Iijima**.



Types Of Carbon Nanotubes



Classified mainly in two types:-

1. Single walled nanotubes
2. Multi walled nanotubes

Other related structures:-

#Torus

#Nanobud

#Graphenated carbon nanotubes (G-CNTs)

#Nitrogen doped carbon nanotubes (N-CNTs)

#Peapod

#Cup-stacked carbon nanotubes

Single wall CNT (SWCNT)

- The structure of a SWNT can be conceptualized by wrapping a one-atom-thick layer of graphite called graphene into a seamless cylinder.
- The way the graphene sheet is wrapped is represented by a pair of indices (n,m) . The integers n and m denote the number of unit vectors along two directions in the honeycomb crystal lattice of graphene.

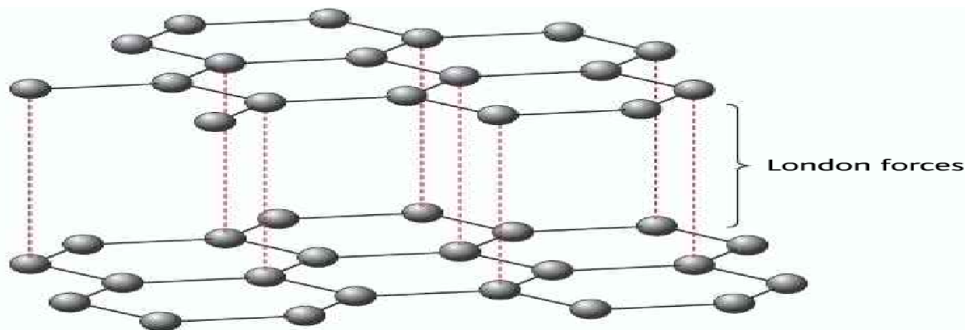
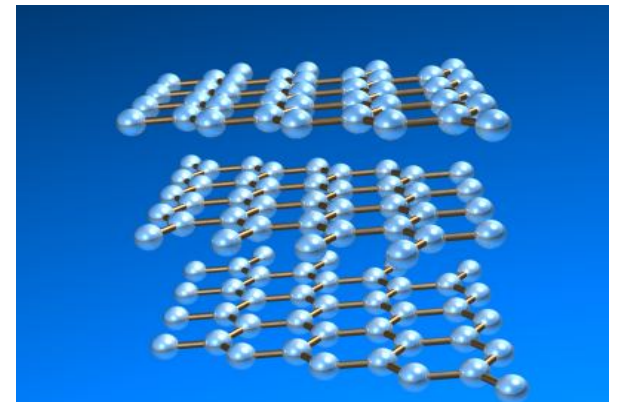


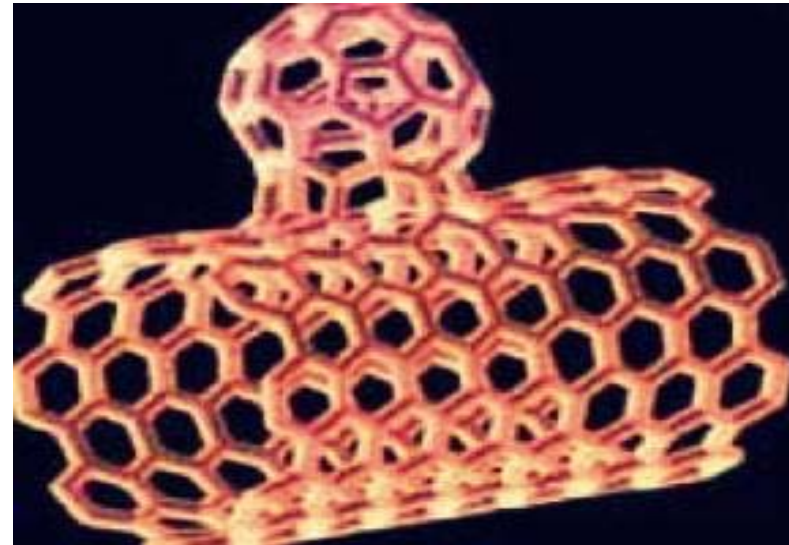
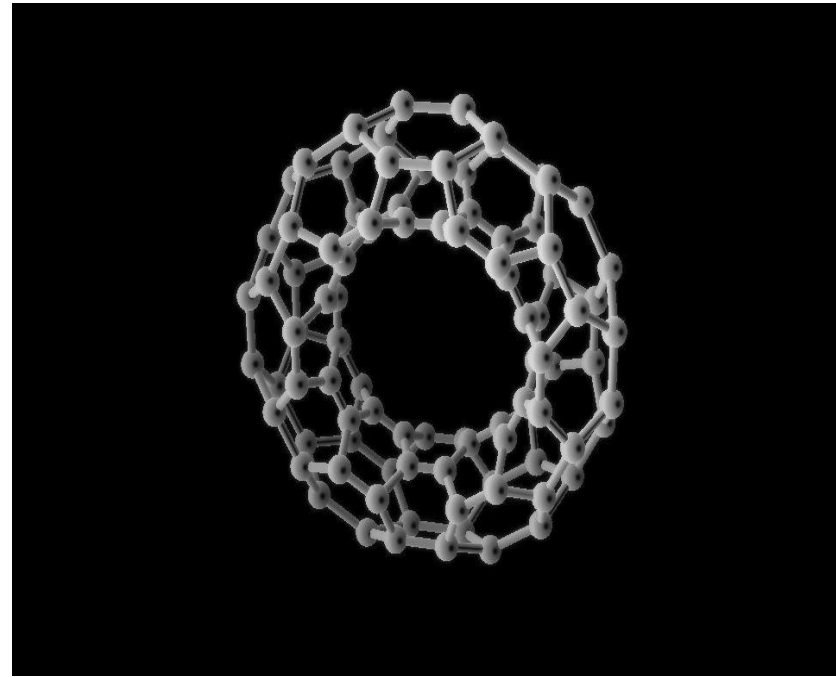
fig. 2.1.17 The structure of graphite. Whilst the bonds between the carbon atoms are strong, those between the layers are not and so they slide over each other easily.



TORUS :-

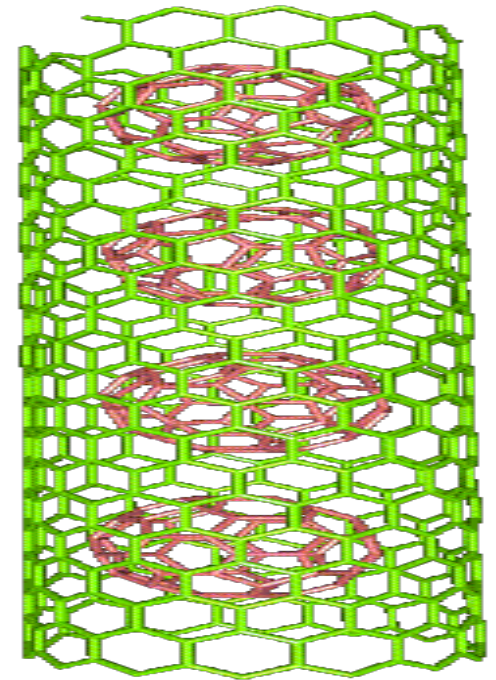
It is a carbon nanotube bent in a torus shape (i.e. doughnut shape)

Carbon Nanobud are created combining carbon nanotubes and fullerenes.



Peapod :-

A carbon peapod is a novel hybrid carbon material which traps fullerene inside a Carbon nanotube.



Cup-stacked carbon nanotubes :-

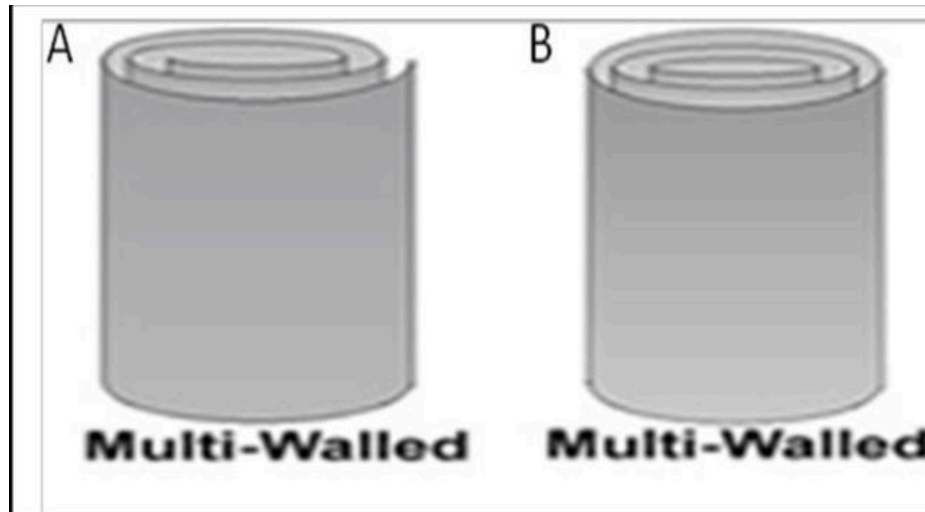
Cup-stacked carbon nanotubes exhibit semiconducting behaviours due to the stacking microstructure of graphene layers.



Multi-walled carbon nanotube (MWNT).

MWNTs consist of multiple rolled layers (concentric tubes) of graphene.

There are two models that can be used to describe the structures of multi-walled nanotubes.



Multi-walled carbon nanotubes (MWCNTs): (a) Parchment model, (b) Russian Doll model

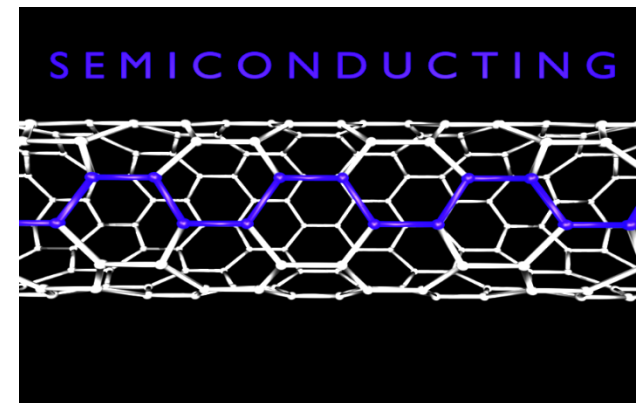
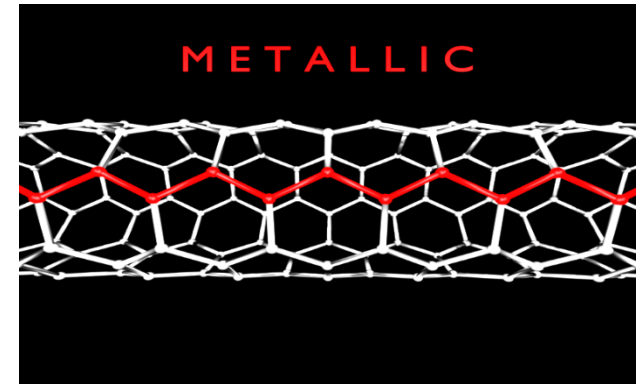
Carbon nanotube properties: mechanical

Orbital hybridization best describes chemical bonding in nano tubes. The chemical bonding of nano tubes is composed entirely of sp^2 carbon-carbon bonds which is stronger than sp^3 carbon-carbon bonds in alkane & diamond. It provides the unique strength of nano tubes.

Material	Young's modulus (GPa)
Steel	190-210
SWNT	1,000+
Diamond	1,050-1,200

Carbon nanotube properties: Electrical

- If the **nanotube structure is armchair** i.e. **$n=m$** : then the electrical properties are metallic. In theory, **metallic nanotubes can carry an electrical current density of 4×10^9 A/cm²** which is more than 1,000 times greater than metals such as copper
- If the **nanotube structure is chiral** : then the electrical properties can be **either semiconducting with a very small band gap**, otherwise the nanotube is a moderate semiconductor



Carbon nanotube properties: Thermal

- The high-frequency carbon-carbon bond vibrations provide an **intrinsic thermal conductivity higher than even diamond**.
- All nanotubes are expected to be **very good thermal conductors along the tube**, exhibiting a property known as "ballistic conduction" but good insulators laterally to the tube axis.
- It is predicted that carbon nanotubes will be able to **transmit up to 6000 watts per meter per kelvin at room temperature**; compare this to **copper**, a metal well-known for its good thermal conductivity, which transmits **385 watts per meter per k**.
- The **temperature stability of carbon nanotubes is estimated to be up to 2800°C in vacuum and about 750°C in air**.

Applications

1 biological and biomedical research

2 composite materials

2.1 mixtures

2.2 textiles

2.3 carbon nanotube springs

2.4 alloys

2.5 coatings and films

3 microelectronics \

3.1 transistors

3.2 thermal management

Applications

- 4 solar cells
- 5 hydrogen storage
 - 5.1 experimental capacity
 - 5.2 limitations on efficient hydrogen adsorption
- 6 electronic components
 - 6.1 interconnects
 - 6.2 electrical cables and wires
- 7 energy storage
 - 7.1 supercapacitor

Applications

- 7.2 batteries
- 8 chemical
- 9 mechanical
- 9.1 actuators
- 9.2 loudspeaker
- 10 optical
- 11 environmental
- 11.1 environmental remediation
- 11.2 water treatment

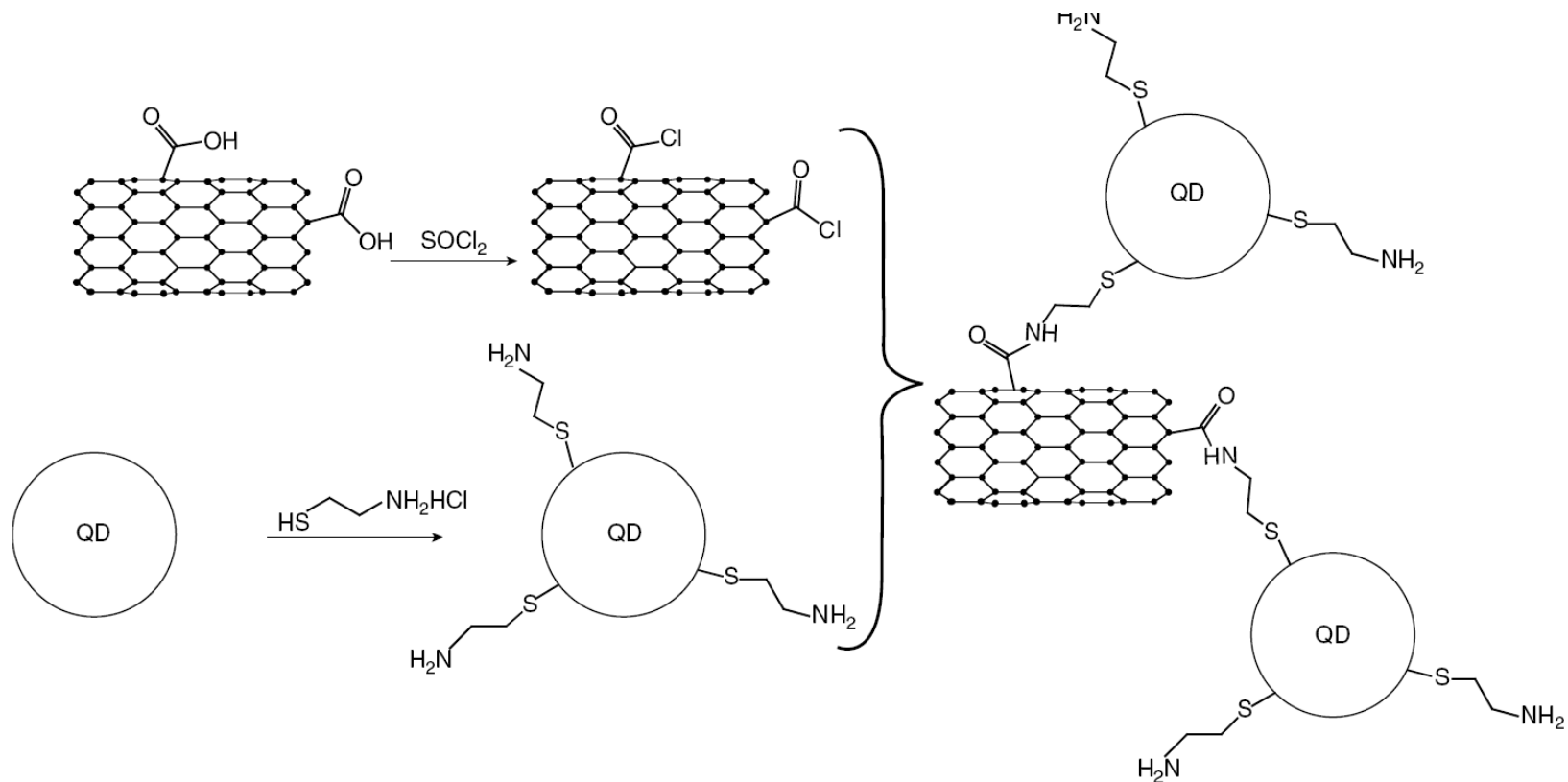


FIGURE 3.18 Quantum dots (QD) attachment at cut ends or at defects along NT sidewall. (From Haremza, J.M. et al., *Nano Lett.*, **2**, 1253–1258, 2002. With permission.)

Environmental , Social & Ethical implications of nanotechnology

Nanotechnology products

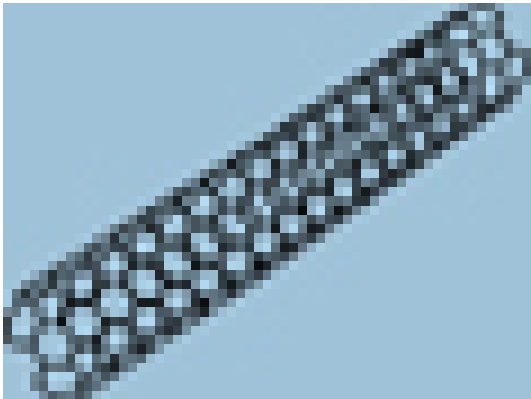
Dermatone
SPF 20
Natural
Formula



Curad Silver
Bandages



NANO B-12
Vitamin Spray

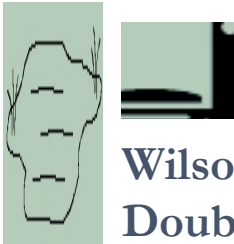


SAMSUNG Samsung Nano
SilverSeal Refrigerator

Eddie Bauer
Ruston Fit
Nano-Tex
Khakis



Wilson
Double-Core
Tennis Balls



Mercedes
CLS Class



Hummer
H2



Areas of Society affected by Nanotechnology

Environment

Health

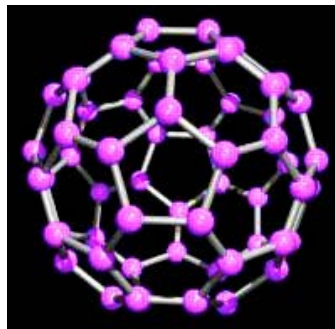
Economy

Possible Benefits

- Improved detection and removal of contaminants
- Development of benign industrial processes and materials
- Improved medicine
- Better products
- New jobs

Nanotechnology challenges

- Do nanomaterials (NM) present new and unique risks for health and safety and for the environment?
- Can the potential benefits of nanotechnology be achieved while minimizing the potential risks?



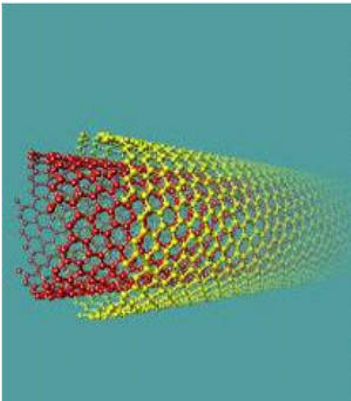
Categories of nanomaterials

- National academy of sciences

- Nanotubes
- Nanoclays
- Quantum dots
- Metal oxides

- EPA

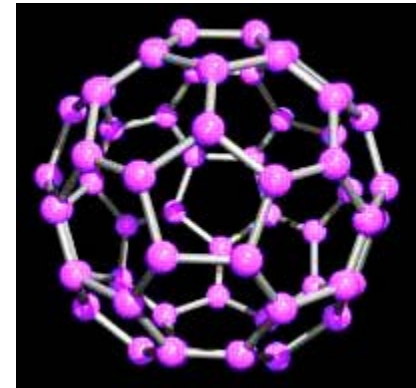
- Carbon-based
- Metal-based
- Dendrimers
- Composites



Carbon Nanotubes

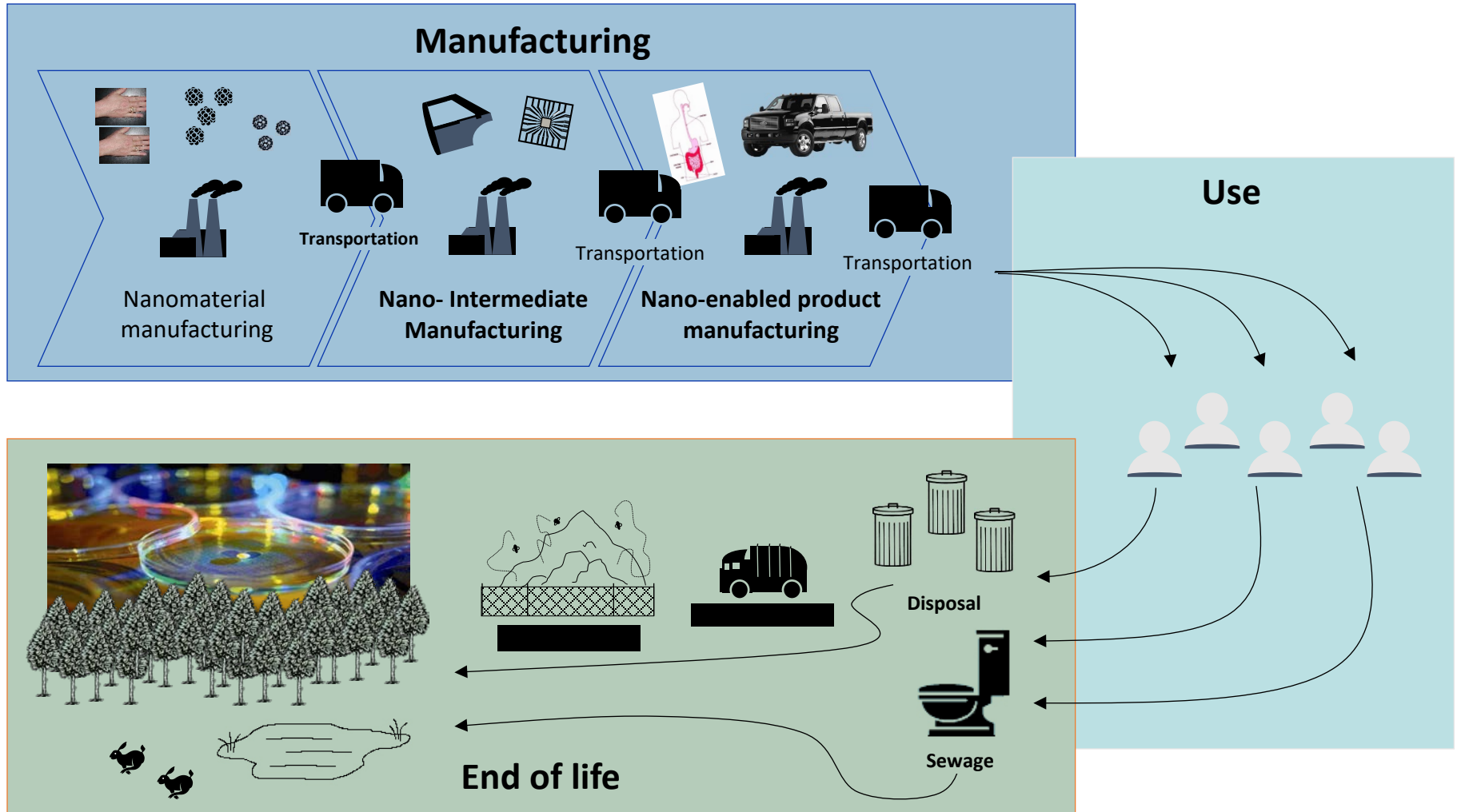


Quantum Dots



Fullerenes

Life Cycle of Nanomaterials



Health, safety, and environmental concerns regarding NM

- **Human implications**

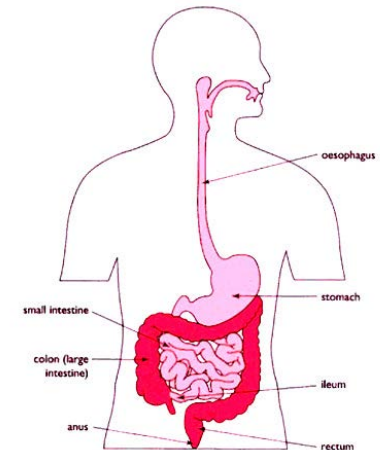
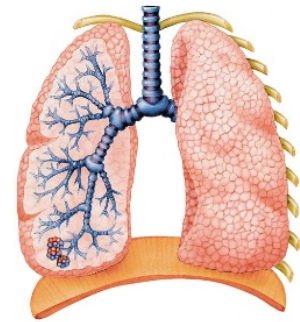
- NM toxicity not yet well understood; nano-size materials do not behave like their bulk counterparts
- Reactivity of NM due to large surface area
- Potential for bioaccumulation

- **Environmental implications**

- Contamination of water and soil from improper disposal of NM
- Bio-uptake of NM and accumulation in food chain

Nanotoxicology

- *Nanotoxicology* – science of engineered nanodevices and nanostructures that deals with their effects in living organisms
- Potential NM exposure routes include:
 - Inhalation
 - Dermal contact
 - Ingestion

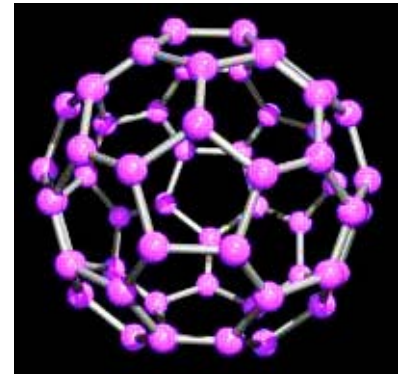


Translocation/bioaccumulation of nanomaterials

- ❖ Nanoparticles can cross alveolar wall into bloodstream
- ❖ Absence of alveolar macrophage response
- ❖ Distribution of NM to other organs and tissues
- ❖ Inhaled nanoparticles may reach brain through olfactory nerve

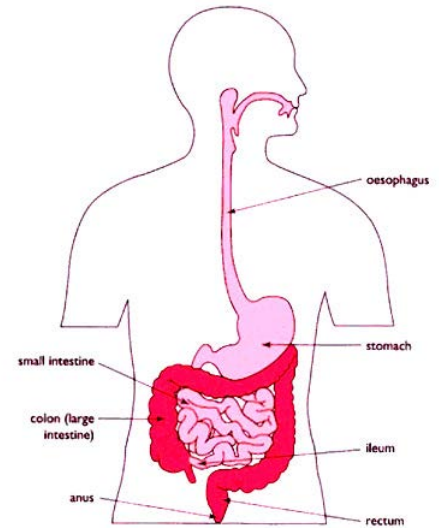
In vitro NM studies

- Fullerenes can interact with cell membranes and specifically with membrane lipids
- Interactions can produce lipid peroxidation and leaky cell membranes that result in the release of cellular enzymes.
- Proposed mechanism of damage is that fullerenes generate superoxide anions



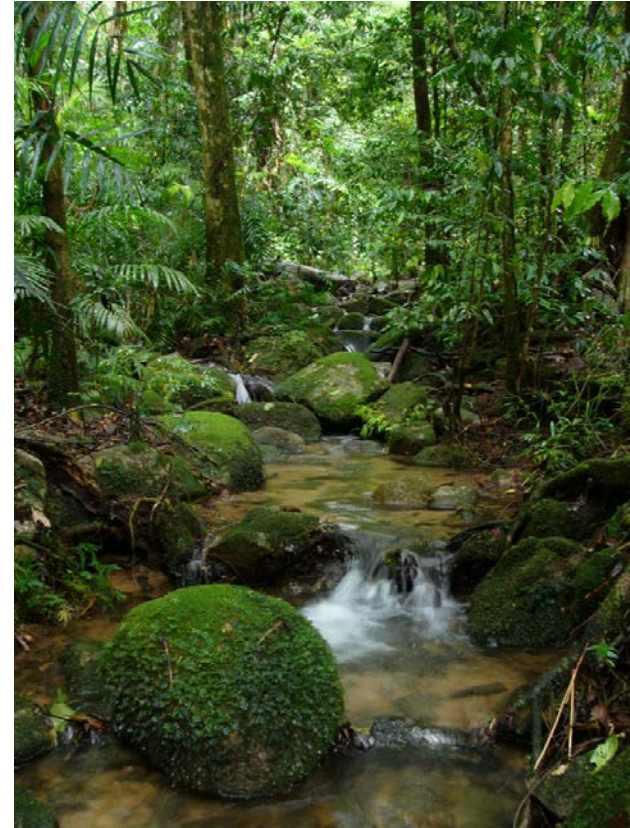
Ingestion pathway

- Ingestion exposures can occur through direct intake of food or materials containing NM and secondary to inhalation or dermal exposures
- Some evidence suggests that ingested NM may pass through to lymphatics
- Little research to date about ingestion exposures and the potential for distribution of NM to other tissues



Environmental risk concerns regarding NM

- ❖ What happens to NM after product use and disposal?
- ❖ What is the fate of NM in the environment?
- ❖ Do NM degrade?
- ❖ Will NM accumulate in the food chain?
- ❖ How to evaluate real world exposures to NM?



Assessing risks of nanomaterials

- Identify and characterize potential NM hazards
- Assess potential exposure scenarios
- Evaluate toxicity
- Characterize risk and uncertainty
- Communicate about risks

Specific Societal and Ethical Issues in Nanotechnology

Lab safety

- consideration of the health and well-being of fellow researchers, reporting on unsafe practices

Environmental consequences of research

- minimization and safe disposal of hazardous substances, fate of “nanowaste,” fair notice to potentially affected parties

Academic conduct

- integrity of research results, equitable authorship recognition practices

Commercial fair dealing

- respect of confidentiality and trade secrets

Science education

- interdisciplinary studies, ethics education

More Societal and Ethical Issues in Nanotechnology

Environmental, health, and safety concerns

- ecological and toxicological effects of nanoparticles, workplace and consumer exposure

Economic effects

- rapid transformation and dislocation of industries, effects on wealth distribution, intellectual property issues

Medical technologies

- prospects for human enhancement and augmentation, improved genetic screening, advanced cures

Security and privacy implications

- novel weaponry and defense technologies, pervasive surveillance potential

Less Obvious Societal and Ethical Issues in Nanotechnology

National and international politics

- national research funding commitments and the “nanodivide,” technology transfer

Media and public perceptions

- polarized reactions to nanotechnology, involvement of lay public in decision making

Legal and regulatory issues

- proactive versus reactive regulation, international standard-setting

Cultural and religious repercussions

- new media and modes of representation, new challenges in defining life

LEGAL IMPLICATIONS OF NANOTECHNOLOGY

Anything affecting society, the economy, medicine, education, the environment and ethics certainly affects the law and legal frameworks. Many legal intersection of nanoscience, nanotechnology, and the law. will become clear as the technology develops, but some predictions have emerged already.

Such implications in

- **Genetics**
- **Legal concepts of property**
- **Intellectual Property Right(IPR)**
- **Privacy**

Liability arises when Nano Technology uses following way

- Military applications in civil war
- Defective nano products in Market place (Warning defect, *Design defect*, Manufacturing defect)
- Malpractice in medicine science
- Using nano-energetic materials that are intended as explosive devices
- Uses of nano-energetic materials for the demolition of structures.
- Unethical Forensic applications of nanotechnology (Fingerprint data hack)
- Using nanotechnology in cyber crime (Hacking, Stealing of Data)
- Unauthorized uses of nano patent, Design, Trademark, Know how, Formula etc.
- Nano-Tracking Devices (For crime, Militant)
- Nano-neural Interventions and Implants(for manipulation of brain processes and, potentially, manipulation of thought)

Challenges of Nanotechnology

- I. Develop Technology
- II. Understand Implications
- III. Prepare Solutions
- IV. Implement Globally

Satisfactory
outcome requires
all four
to be effectively
addressed.